

DAY 5

Day5: Troubleshooting Network Services Practical (1 hour): Troubleshoot DNS and DHCP issues in a simulated network. Task: Document solutions for common DNS and DHCP issues.

[Q-1] Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

Aim: To simulate a DNS resolution failure by editing the hosts file and observing the error.

Procedure

1. Open Notepad as Administrator.
2. Open the hosts file located at:

C:\Windows\System32\drivers\etc\hosts

3. Add the following entry:

1.1.1.1 www.spotify.com

4. Save the file.
5. Open Command Prompt.
6. Run the following command:

ping www.spotify.com

Observed Output

Pinging www.spotify.com [1.1.1.1] with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 1.1.1.1:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Result

The hosts file redirected www.spotify.com to an incorrect IP address. As a result, DNS resolution failed and the ping request could not reach the correct server.

[Q-2] Configure DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue. Aim: To configure a DHCP server with an incorrect default gateway and troubleshoot the issue.

Configuration

- Network: 192.168.1.0/24
- DHCP Address Pool: 192.168.1.10 – 192.168.1.100
- Incorrect Default Gateway: 192.168.1.254
- Correct Default Gateway: 192.168.1.1

Verification

On the client PC, execute:

```
ipconfig /all
```

Assigned Configuration

IPv4 Address : 192.168.1.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.254

Symptoms

- Internet access is unavailable.
- Local network communication may still work.
- The assigned gateway cannot be reached.

Troubleshooting

1. Run ipconfig /all to verify the assigned gateway.
2. Ping the default gateway.
3. Check the DHCP server configuration.
4. Compare the assigned gateway with the router's actual IP address.

Solution

1. Open the DHCP server configuration.
2. Change the default gateway from 192.168.1.254 to 192.168.1.1.
3. Renew the client IP configuration using:
4. ipconfig /release ipconfig /renew

Result

The client received the correct default gateway and network connectivity was restored.

[Q-3] Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Scenario

Devices can access websites using IP addresses but cannot access them using domain names.

Diagnostic Command

nslookup google.com

Sample Output

DNS request timed out.

Server: Unknown

Request timed out.

Troubleshooting Steps

5. Verify network connectivity.

6. Ping a public IP address such as:

ping 8.8.8.8

7. Run:

nslookup google.com

8. Check whether a valid DNS server is configured.

9. Flush the DNS cache:

ipconfig /flushdns

10. Restart the DNS Client service if required.

11. Configure a valid DNS server such as 8.8.8.8 or 8.8.4.4.

Solution

The issue was caused by an incorrect or unreachable DNS server. After configuring a valid DNS server, domain names were resolved successfully.

[Q-4] A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

Problem

The client receives an APIPA address instead of an IP address from the DHCP server.

Example:

169.254.25.10

Possible Causes

- DHCP server is powered off.
- Network cable is disconnected.
- Switch port is disabled.
- DHCP service is not running.
- DHCP address pool is exhausted.
- Incorrect VLAN configuration.
- Firewall is blocking DHCP traffic.

Resolution Steps

12. Check all network cable connections.
13. Verify that the network interface is enabled.
14. Ensure the DHCP server is running.
15. Check whether the DHCP scope has available IP addresses.
16. Verify VLAN configuration.
17. Release and renew the IP address:
ipconfig /release ipconfig /renew
18. Restart the DHCP service if necessary.
19. Confirm that the client receives a valid IP address.

Result

After correcting the DHCP issue, the client obtained a valid IP address from the DHCP server.

[Q-5] Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Common DNS and DHCP Issues

Service	Issue	Symptoms	Possible Cause	Recommended Solution
DNS	Incorrect DNS Server	Websites open by IP but not by name	Wrong DNS server configuration	Configure a valid DNS server address
DNS	Incorrect Hosts File Entry	Specific website is unreachable	Wrong IP mapping in the hosts file	Remove or correct the hosts file entry
DNS	DNS Server Unavailable	DNS lookup fails	DNS server is down or unreachable	Restart or replace the DNS server
DHCP	Incorrect Default Gateway	Internet access unavailable	Wrong gateway configured in DHCP	Configure the correct default gateway
DHCP	DHCP Scope Exhaustion	No IP address assigned	DHCP address pool is full	Expand the DHCP scope or remove unused leases
DHCP	DHCP Server Unavailable	Client receives APIPA (169.x.x.x) address	DHCP service stopped or server unreachable	Start the DHCP service and verify network connectivity